

Determining Role of Largemouth Bass Virus in Mortality of Juvenile Smallmouth Bass

Deciphering the Role of the Largemouth Bass Virus (LMBV) in Causing the Observed Mortalities in Juvenile Smallmouth Bass Micropterus Dolomieu in the Susquehanna River Basin, Pennsylvania

Since 2005, mortality episodes of young-of-year (YOY) Smallmouth Bass *Micropterus dolomieu* (SMB) were consistently reported from a number of river systems in Pennsylvania, such as the Susquehanna, Juniata, and Allegheny rivers. These mortality events were more persistent and severe in the Susquehanna and Juniata River systems, occurring annually to varying degrees, than in the Allegheny and other drainages. These SMB die-offs have created considerable concerns among the sport-fishing industry, as well as in state and federal agencies, as the relative abundance of YOY and adult SMB decreased and shifts in size structure were noticed (Smith et al., 2015).

Several different bacterial pathogens, myxozoan and trematodes parasites, and the viral pathogen Largemouth Bass Virus (LMBV) have been isolated from moribund SMB over the course of the investigation. LMBV has consistently been isolated from moribund YOY SMB from the Susquehanna River Basin; however, the role that LMBV may play in causing the underlying stress is unknown. Previous studies have not been designed to determine if LMBV is pathogenic to SMB, but its consistent detection in moribund YOY SMB warranted further investigation. This study evaluated whether LMBV isolates from Susquehanna River Basin are pathogenic to YOY Smallmouth Bass and at what concentration (i.e., LD50) in a laboratory setting. Further, we evaluated the role that temperature regimes similar to those observed in the focus area and co-occurring opportunistic bacteria known to be pathogenic to SMB had in inducing clinical disease in YOY Smallmouth Bass

First, we screened five LMBV strains that were associated with YOY SMB mortality episodes for their pathogenicity to SMB using an intraperitoneal (IP) injection method. Second, we established the median lethal dose (LD50) by immersion of the two most pathogenic LMBV strains. Third, we described gross clinical signs and histopathological alterations of LMBV-experimentally infected fish. Fourth, we compared the five LMBV strains phylogenetically using major capsid protein (MCP) gene sequencing. Fifth, we determined the optimal temperature for LMBV to cause disease in SMB by immersion. Last, we determined the role that two other fish-pathogenic bacteria may play in exacerbating the LMBV infection. The series of experiments described herein unraveled some of the potential mechanisms leading to the YOY SMB mortality episodes noticed in several watersheds in Pennsylvania.

Exposure of SMB to LMBV by IP resulted in morbidity and mortality in all LMBV infected groups with variable LD50 levels that ranged from as little as <10 TCID50/fish for the 13-295 Susquehanna strain to 103.13 TCID50/fish in the case of the 14-204 Pine Creek strain. In LMBV immersion studies, gross clinical signs in LMBV-infected SMB were, to a great extent, similar to those observed in IP-infected fish; however, there were some lesions in the two immersion groups that were not observed in the IP infected SMB groups. When the water temperature was maintained at 23 °C, cumulative mortality was 10% and occurred on the 13th day post-infection. However, when water temperatures were maintained at 28 °C, half of the fish died. During co-infection challenge studies, when LMBV-infected SMB were exposed to *F. columnare*, a significant ($P < 0.05$) increase in cumulative mortality was observed ($75.0 \pm 12.5\%$). SMB exposed to *A. salmonicida* only experienced a very high cumulative mortality of $95.8 \pm 7.2\%$ by 16 days post infection (PI). Mortality of co-challenged SMB began 10 days PI and continued until 27 days PI, reaching a cumulative mortality of 100%.

The findings of this study clearly demonstrate that LMBV infection, regardless of the method of exposure, is lethal to YOY SMB. The variability in LD50 values among the five SMB LMBV strains can be attributed to inherent variability in pathogenicity among LMBV strains, a phenomenon that is common in most animal viruses, including LMBV (Grgić et al., 2008; Kang et al., 2014). Based on the gross and microscopic lesions, it is possible that the dermal lesions observed in SMB in affected rivers are initiated primarily by LMBV and that affected areas became colonized by opportunistic bacteria and fungi; both of which are abundant in the aquatic environment during the summer. Similar to LMB, water temperature plays an important role in LMBV pathogenicity for SMB under controlled experimental conditions. Raising the water temperature above 23 °C was necessary for successful experimental infection of SMB with LMBV. This coincides well with the water temperatures recorded during the YOY SMB mortality episodes that fluctuated between 22 and 34 °C. High water temperatures are often associated with other stressors including low dissolved oxygen concentrations. These factors favor LMBV replication and can compromise host defense mechanisms. Additionally, co-infection of LMBV with *F. columnare* or *A. salmonicida* may have important implications for YOY SMB mortality across a wide temperature range. During warmer summer months, LMBV alone or in combination with *F. columnare* can cause YOY SMB mortality at the magnitude noticed in the multiple river systems. *F. columnare* alone cannot be the primary cause of mortality since, even at extremely high concentrations in experimental settings, it does not cause mortality levels as high as reported for the Susquehanna and Potomac River basins.

The sum of data generated in this study, including LMBV ability to cause, without co-infection, high mortality rates associated with dermal lesions under laboratory conditions that are visually similar to the skin lesions observed in YOY SMB during the mortality episodes, and its relatively high optimal temperature which coincides with those prevailing at the affected rivers during the peak of mortality episodes, strongly suggest that this iridovirus is the most likely primary cause of this large-scale mortality afflicting SMB. While chemicals and other adverse environmental factors could be indirectly involved (e.g., immune suppressor), their role in the mortality episodes remains unclear.

Publications:

Boonthai, T., Lock, T.P., Yamashita, C.J., Smith, G.D., Winters, A.D., Kiupel, M., Brenden, T.O., and Faisal, M. 2018. Laboratory investigation into the role of largemouth bass virus (Ranavirus, Iridoviridae) in smallmouth bass mortality events in Pennsylvania Rivers. BMC Veterinary Research 14: 62.

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Pennsylvania Fish and Boat Commission

Funding Amount:

\$50,114

Research Year:

2014

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